Customer Shopping Behavior Analysis: End-to-End Data Project Report

1. Executive Summary

This project presents an end-to-end data analytics workflow on a retail dataset comprising **3,900 shopper transactions** representing **$233,081 in total revenue**. The primary goal of this investigation was to identify hidden revenue opportunities, optimize marketing and promotional discount strategies, evaluate subscription retention mechanics, and establish actionable recommendations to convert single-time transaction buyers into loyal brand advocates.

Using a multi-stage technical pipeline combining **Python (Pandas, NumPy)** for cleaning and feature engineering, **MySQL** for database warehousing and complex querying (CTEs, Window Functions, and aggregations), and PowerPoint for executive visual presentation, key operational insights were uncovered:

- **Demographic Skew:** Male customers generate **67.7% of gross revenue** ($157,890 across 2,652 orders), while female shoppers account for **32.3%** ($75,191 across 1,248 orders).
- **Customer Lifetime Value & Loyalty: 79.9%** (3,116) of shoppers fall into the "Loyal" segment (>10 purchases), but new customer acquisition sits at a critical deficit of **2.1%** (83 first-time buyers).
- **Subscription Program Performance:** Active subscribers do not exhibit higher average order values ($59.49 vs. $59.87 for non-subscribers). Subscription value lies in repeat order frequency and customer retention.
- **Discount & Logistics Dynamics:** 839 customers applied promotional discounts while spending above the storewide average ($59.76). Express shipping users spent 3.5% more per transaction ($60.48 vs. $58.46).

2. Technical Stack & Implementation Workflow

Data Cleaning & Feature Engineering (Python / Jupyter Notebook)

1. **Missing Value Imputation:** Handled missing data in `review_rating` (37 missing values) by imputing category-level medians rather than overall column averages to preserve item category distribution integrity.
2. **Standardization & Casing:** Applied snake_case naming conventions across all 18 attributes for smooth SQL database ingestion.
3. **Feature Creation:**
    - `age_group`: Categorized shopper ages into quartile-based brackets (`Young Adult`, `Adult`, `Middle-aged`, `Senior`).
    - `purchase_frequency_days`: Mapped text frequency descriptors into standardized numeric day equivalents (`Weekly` = 7, `Fortnightly` = 14, `Monthly` = 30, `Quarterly` = 90, `Annually` = 365).
4. **Data Redundancy Reduction:** Dropped the redundant `promo_code_used` feature after validating 100% correlation with `discount_applied`.

Database Warehousing & SQL Querying (MySQL)

The processed dataset was exported directly to a local MySQL instance using `SQLAlchemy` and `PyMySQL`. Ten analytical SQL queries were developed to answer core business questions:

- **Revenue Aggregations:** Evaluated revenue contributions by gender, age groups, and shipping methods.
- **Filtering & Subqueries:** Identified high-value shoppers using promotional discounts above average spend thresholds.
- **Window Functions & Ranking:** Utilized `ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC)` to isolate top-3 purchased items per category.
- **Customer Segmentation (CTEs):** Segmented buyers into `New` (1 purchase), `Returning` (2–10 purchases), and `Loyal` (>10 purchases) categories.

3. Key Findings & Analytics Insights

A. Shopper Demographics & Age Analysis

Gender	Total Revenue	Transactions	Revenue Share	Avg Basket Size
Male	$157,890	2,652	67.7%	$59.54
Female	$75,191	1,248	32.3%	$60.25

Insight: Average basket size is nearly identical across genders; the revenue gap is purely driven by overall customer transaction volume.

Age Bracket	Age Range	Total Revenue	Shopper Count	Spend per Shopper
Middle-aged	46–60	$67,711	1,135	$59.66
Adult	31–45	$64,927	1,093	$59.40
Young Adult	18–30	$57,279	949	$60.36
Senior	60+	$43,164	723	$59.70

B. Customer Segmentation & Subscription Analysis

Segment Tier	Definition	Shopper Count	Percentage
Loyal Buyers	>10 Purchases	3,116	79.9%
Returning Buyers	2–10 Purchases	701	18.0%
First-Time Buyers	1 Purchase	83	2.1%

Subscription Status	Shopper Count	Avg Transaction Spend	Operational Impact
Non-Subscribers	2,847	$59.87	Baseline single-order margin

Subscription Status	Shopper Count	Avg Transaction Spend	Operational Impact
Active Subscribers	1,053	$59.49	High order frequency & lower churn

Takeaway: Subscriptions do not drive higher order values directly; rather, they secure predictable, recurring purchase frequency across the calendar year.

C. Merchandise & Pricing Dynamics

Category	Top 3 Performing Items	Order Counts
Accessories	Jewelry, Belt/Sunglasses, Scarf	171, 161, 157
Clothing	Blouse/Pants, Shirt, Dress	171, 169, 166
Footwear	Sandals, Shoes, Sneakers	160, 150, 145
Outerwear	Jacket, Coat	163, 161

- **Discount Penetration:** Items exhibiting the highest discount reliance include Hats (50.0%), Sneakers (49.7%), Coats (49.1%), and Sweaters (48.2%).

4. Strategic Business Roadmap

1. **Target Female Shopper Acquisition:** Expand digital ad spend and marketing campaigns tailored to female demographics to rebalance the current 68/32 revenue skew.
2. **Implement Smart Discount Thresholds:** Replace blanket promo codes with minimum spend requirements (e.g., "$15 off on orders over $80") to protect margins on high-demand items.
3. **Revamp Subscription Program Value Proposition:** Introduce exclusive subscriber perks, early access to new collections, and free express shipping rather than relying on order discounts.

4. **Solve the Top-of-Funnel Acquisition Bottleneck:** Launch targeted first-time buyer welcome offers and onboarding campaigns to convert new visitors and fix the low 2.1% new customer conversion rate.

5. Project Deliverables & Repository Structure

- **Python Notebook:** Data cleaning, missing value imputation, and database ingestion (`Customer shopping behavior analysis.ipynb`).
- **SQL Script:** Complete MySQL analytical script (`custmoer_behaviour_mysql_queries.sql`).
- **Executive Presentation:** 10-slide deck with visual charts (`Customer Shopping Behavior Analysis.pptx`).